

# Tyler Roberts

Senior Software Engineer

## CONTACT

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## SKILLS

### Programming

C#, .NET, .NET Core, ASP.NET Core

### Data & ORM

Entity Framework Core, LINQ, System.Linq.Expressions, Expression Trees, EF Core SQL Translation, Query Providers

### Advanced C#

ExpressionVisitor, Generics, Delegates, Higher-Order Functions, Reflection, Runtime Type Generation

### Architecture & Patterns

Fluent API Design, Builder Pattern, Domain-Driven Design, Extensible Framework Design

### Databases

SQL Server, PostgreSQL, Relational Database Design

### Cloud & DevOps

AWS, CI/CD, Docker, Observability

## EDUCATION

### Florida State University

Bachelor of Computer Science  
2009 - 2013

## Professional Summary

Senior Software Engineer with 9+ years of experience designing and building production-grade .NET applications with deep expertise in LINQ, Entity Framework Core, and System.Linq.Expressions. Specialized in constructing and rewriting expression trees at runtime using ExpressionVisitor to enable dynamic query composition, fluent APIs, and extensible data engines. Strong understanding of EF Core internals, SQL translation pipelines, and query optimization, with proven ability to refactor complex C# logic into SQL-compatible projections and joins. Experienced in advanced metaprogramming, runtime type generation, and fluent builder patterns to support highly configurable and scalable enterprise systems. Adept at owning core platform engines end-to-end, focusing on correctness, performance, maintainability, and extensibility.

## Professional Skills

### Senior Software Engineer

Web Venturez LLC | February 2020 - October 2025

- Designed and implemented core .NET application engines with heavy use of LINQ and Entity Framework Core to support complex, data-driven enterprise workflows.
- Built dynamic query frameworks using System.Linq.Expressions, manually constructing and rewriting expression trees at runtime via ExpressionVisitor.
- Developed fluent APIs and builder-style configuration layers enabling consumers to define queries, filters, and projections through expressive C# syntax.
- Worked deeply with EF Core internals, diagnosing and resolving SQL translation issues by restructuring LINQ expressions into SQL-compatible joins and projections.
- Implemented custom abstractions over IQueryable to enable composable, reusable, and testable query logic across services.
- Used advanced generics, delegates, and higher-order functions to treat business logic as data and enable runtime extensibility.
- Implemented runtime type generation and projection shaping to handle variable data models without introducing memory leaks.
- Optimized query performance by analyzing generated SQL and refactoring expression trees to reduce complexity and execution cost.
- Collaborated with product and architecture stakeholders to translate ambiguous requirements into robust, extensible platform capabilities.

- Introduced structured logging and diagnostics around query generation to improve observability and troubleshooting of dynamic queries.
- Owned critical platform components end-to-end, ensuring correctness, scalability, and long-term maintainability of the data access layer.

## **Full Stack Engineer**

Webiz Digital | July 2017 - January 2020

- Developed .NET-based backend services using Entity Framework and LINQ to support complex business workflows.
- Built reusable query components leveraging expression trees to dynamically compose filters and projections.
- Designed fluent configuration APIs that abstracted data access complexity from higher-level application logic.
- Worked extensively with EF Core to troubleshoot translation errors and refactor logic into efficient SQL queries.
- Applied advanced C# generics and delegates to create flexible, reusable infrastructure components.
- Collaborated closely with frontend and product teams to align backend query capabilities with UI requirements.
- Optimized database access patterns by analyzing LINQ-to-SQL output and eliminating inefficient queries.
- Contributed to architectural decisions around extensibility, maintainability, and performance.
- Improved code quality through refactoring, testing, and enforcing consistent query-building patterns.

## **Software Developer**

Centric Tech Inc. | October 2013 - June 2017

- Built and maintained .NET applications using LINQ and Entity Framework for data-intensive systems.
  - Implemented dynamic filtering and sorting logic using expression trees and generic query helpers.
  - Assisted in designing reusable data access layers that supported evolving business requirements.
  - Worked with senior engineers to understand query translation, performance tuning, and SQL generation.
  - Refactored legacy data access code into cleaner, more maintainable LINQ-based implementations.
  - Supported debugging and optimization of complex queries in production environments.
  - Established a strong foundation in advanced C# patterns, data access design, and enterprise application development.
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